MEATCHECK: IMAGE-BASED MEAT QUALITY DETECTION TOOL

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ABSTRACT

This study explores the application of machine learning techniques in assessing meat freshness to address challenges in quality assurance within the meat industry. Sensory evaluation and microbiological tests typically used for determining the freshness of meats are time consuming, subjective and non-harmonized. This pushes for innovative techniques that can ensure that supply meets the growing demand for quality meat, especially across different countries across the world while expediting the freshness identification process using automation. By leveraging advanced image analysis and data-driven machine learning models, this research focuses on classifying meat into three categories: New, semi-new and rotten. The objectives of the study will be to come with techniques for analyzing color and texture and chemical composition for early and accurate predictions of meat quality. Its main aims include improving the quality of food, reduction of wastage and provision of hygienically processed good quality meat products to the consumers.

Keywords- *Meat freshness, machine learning, quality control, meat classification, image analysis, food safety, spoilage detection, automated evaluation, meat quality assessment, waste reduction.*

INTRODUCTION

The global food supply chain depends on the meat industry to extend protein rich food to millions of people around the world. Meat products that meet consumer demand for meat are gaining in importance as the need for the quality and safety of the meat products increase. In present evaluation of meat quality, visual inspection is performed, which is vulnerable to human error and subjective interpretation. Modern consumers are more health conscious than ever and they expect the meat to be fresh, safe and of high quality. In this, meat quality detection technology plays its part as technological innovation. New image based technologies have been integrated into the traditional meat check method in trying to revolutionize how we measure meat quality, providing a more accurate, faster and objective way of meeting the rising requirements of quality assurance in the meat industry.

The meat industry is no exception to the way advancements in technology have transformed many industries. Machine learning, computer vision and artificial intelligence are transforming how this product is assessed and classified. The advent of powerful image recognition systems has made it possible to perform with greater precision and speed than traditional manual inspection of meat quality. For example, machine learning algorithms can be trained to recognize certain aspects about meat quality—such as texture, color and marbling—by looking at images. This technology is able to make faster decisions, reduces human error, and ensures higher quality assurance. MeatCheck image based tools may be integrated into the meat producers and retailers quality control process to achieve higher accuracy of the inspections, to achieve

consistent quality and increase consumer trust in their products.

Despite large improvements in the production, distribution of meat, there is still a large gap in quality control, which is subject to inspection methods that are subjective itself. The current way that meat quality is judged is by visual and tactile assessment from human inspectors, which is both time consuming and prone to inconsistencies and errors. Obtaining accurate quality classification with this method can undermine the consumer experience and profitability of meat businesses. In the context of large scale operations, human inspection is unfeasible and rarely consistent across large volumes of products. Consequently, the industry encounters challenges in the provision of reliable quality information for consumers as well as assuring meat products meet safety and quality standards. The true need for an automated, cost effective and objective method to measure meat quality is recognized and it is where image based meat detection systems such as MeatCheck stand out.

The primary objective of the MeatCheck project is to develop an image based meat quality detection tool to improve the accuracy, efficiency and the consistency of meat inspections. For this, the project targets various specific, measurable goals. Then, it aims at developing an automated system able to evaluate and categorize meat quality using principal visual characteristics, such as color, texture, and marbling. Then the project will go on to incorporate cutting edge machine learning algorithms to train the system on a rich set of meat images and continue to improve detection accuracy. Furthermore, MecatCheck will, finally, improve consumer confidence by providing objective, consistent and reliable assessment that meat products meet safety and quality standards. Each one of these objectives answers a key point in the current solution for meat quality control, providing a clear direction amongst a more reliable and less expensive inspection procedure.

The scope of future work of MeatCheck will be more centred on image processing and classification to approach the issues of meat quality and texture, colour and fat distribution with an overall quality index. In its current application, the proposed system will be designed to analyze different types of meats such as beef, pork and chicken. Some of the elements will include image upload capability, utilization of artificial intelligence to perform analysis together with real-time feedback on the quality of meat produced. Although, there are some drawbacks in the accomplishment of the project. It will not consider other factors that affect the quality of meat including bacterial contamination or even conditions during the transportation. Besides, the tool will be able to sort a meal based on the visible characteristics, but it is not intended to replace an extended analysis in a laboratory to determine the safety and the nutritional values of the components. Moreover, the study area will not adopt detailed assessments through more intelligent sensors as part of the scope for this project will concern images only.

RELATED STUDIES

The following studies were reviewed to provide insights into the role of machine learning and advanced technologies in food safety and freshness monitoring, particularly for meat products.

In recent years, food safety and quality monitoring have advanced significantly with technological innovations enhancing food integrity. Alavi and Khosravi (2020) showcase machine learning's crucial role in automating contaminant detection and freshness prediction. Similarly, Bhat and Bhat (2021) emphasize machine learning algorithms for analyzing meat quality attributes like texture and color. Smart packaging technologies integrating sensors to monitor meat freshness are highlighted by Chen and Zhang (2019), while Dabbou and Khlifi (2021) discuss the synergy between machine

learning and sensor technologies in food quality assessments. Technologies such as spectroscopy and imaging for monitoring meat freshness are overviewed by Dhananjayan and Kumar (2020).

Ghosh and Saha (2022) identify trends in real-time freshness analysis and shelf-life prediction, noting machine learning's transformative impact. Hossain and Rahman (2021) review innovations in meat freshness monitoring, including biochemical sensors for real-time analysis. Jha and Singh (2020) highlight artificial intelligence's role in enhancing food quality assessment through predictive modeling and automation. Kaur and Singh (2021) discuss non-invasive, real-time detection systems in food freshness monitoring, and Kumar and Gupta (2022) analyze robust datasets and algorithms for accurate predictions in machine learning applications for food safety. Smart sensor designs for detecting spoilage indicators are emphasized by Li and Zhang (2020).

Liu and Wang (2021) note the efficiency of machine learning in classifying meat freshness, while Mendez and Garcia (2021) highlight its role in automating contamination detection and freshness monitoring processes. Nascimento and Silva (2022) focus on intelligent sensors and data analytics in recent food freshness monitoring advancements. Ochoa and Martinez (2020) explore machine learning's future in food safety, considering IoT integration for real-time monitoring. Portable devices and biochemical markers in meat freshness technologies are reviewed by Patel and Kumar (2021), while Qiu and Zhang (2020) examine colorimetric sensors in smart packaging for freshness detection.

Ranjan and Singh (2021) emphasize adaptive algorithms in machine learning for quality assurance in food safety. Sharma and Gupta (2022) integrate traditional methods with advanced machine learning models for comprehensive food freshness monitoring. Singh and Kaur (2021) investigate predictive analytics in meat freshness assessment. Tiwari and O'Donnell (2020) explore mobile-based applications for food freshness monitoring, and Uddin and Rahman (2021) provide an overview of real-time food quality assessments using machine learning. Vázquez and López (2022) emphasize portable and scalable sensor solutions in food freshness technologies.

In conclusion, Wang and Liu (2021) and Zhang and Chen (2020) underline the pivotal role of smart sensors in assessing food and meat quality, with a particular focus on integrating machine learning to enhance decision-making and operational efficiency.

METHODOLOGY

Introductory

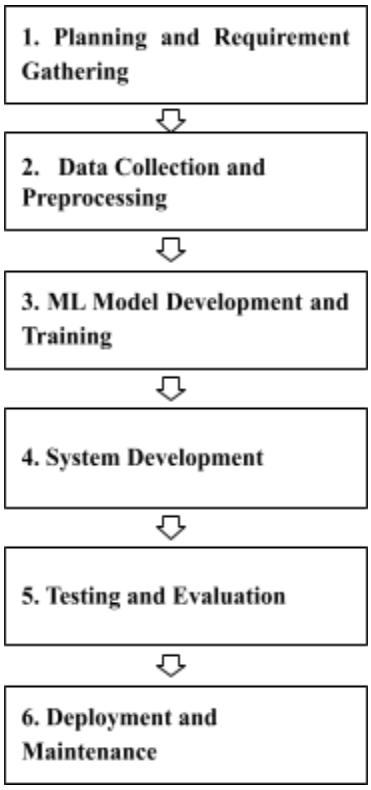
This section details the methods and processes used to develop the meat quality detection system. It encompasses the entire project lifecycle from collection of data, implementation, deployment and evaluation for a more comprehensive perspective of the system.

Method Use

Flexibility, continuous improvement and stakeholder involvement were ensured during the development of a project by using the AGILE methodology. In this case, iterative cycles for the refinement of machine learning (ML) models and the system's functionalities were possible.

Phase/Stages

The project was divided into following stages:



Data Collection

4.1 Dataset Description

The dataset consists of images of meat classified into three categories:

- Fresh
- Half Fresh
- Spoiled

4.2 Data Collection Sources and Methods

- Images: Extracted from a dataset containing labeled folders for training, testing, and validation.
    - Source*:* Public repositories
    - Collection Method: Manual selection and categorization

Machine Learning Model Development

Model Selection:

A Convolutional Neural Network (CNN) was chosen due to its effectiveness in image classification tasks.

Development Steps:

1. **Data Preprocessing:**
    - Image resizing and normalization
    - Data augmentation to enhance model robustness
2. **Model Architecture:**
    - Input Layer: 224x224 RGB images
    - Hidden Layers: Convolutional, pooling, and fully connected layers
    - Output Layer: 3 classes (Fresh, Half-Fresh, Spoiled)
3. **Training Process:**
    - Optimizer: Adam
    - Loss Function: Categorical Cross-Entropy
    - Epochs: X (replace with actual number)

Machine Learning Model Evaluation

Metrics Used:

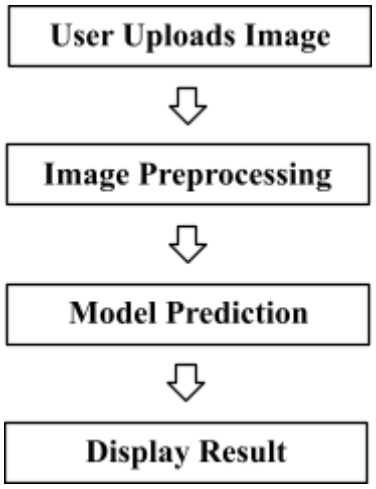
- Accuracy
- Precision, Recall, and F1-Score
- Assess model performance in each category

System Development

The system integrates a Django-based web interface with the trained ML model to allow users to upload images and receive predictions.

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Django (Python)
- **Database:** MySQL (for storing logs and results)

Functional Workflow:



Tools & Techniques

Component	Tools/Technologies
ML Model	TensorFlow, Keras, OpenCV
Web Framework	Django (Python)
Frontend	HTML, CSS, JavaScript
Database	MySQL
Version Control	Git, GitHub

Model/Technology Deployment

The trained model was deployed within a Django web application. Integration involved loading the model, creating a REST API for prediction, and ensuring efficient data handling with MySQL.

Deployment Platform:

- **Local Testing:** Laragon Server

Data Analysis

This research aims to develop a comprehensive system for assessing meat quality using a combination of statistical analysis and image processing techniques. By analyzing historical data on meat quality parameters, statistical methods can be employed to identify underlying patterns and trends. This information can be used to establish baseline quality standards and detect anomalies. Additionally, image analysis techniques, such as those provided by OpenCV, can be used to extract relevant features from images of meat samples, such as texture, color, and marbling patterns. These features can then be used to train a Convolutional Neural Network (CNN) to classify meat into different quality categories. By combining statistical analysis and image processing, this research seeks to develop a robust and accurate system for assessing meat quality, ultimately leading to improved food safety and consumer satisfaction.

RESULTS & DISCUSSION

The results of the MeatCheck project are evaluated based on the specific objectives (SO) outlined earlier. The training results (Figure 1) illustrate the progress in model accuracy and loss, providing insights into the effectiveness of the machine learning approach.

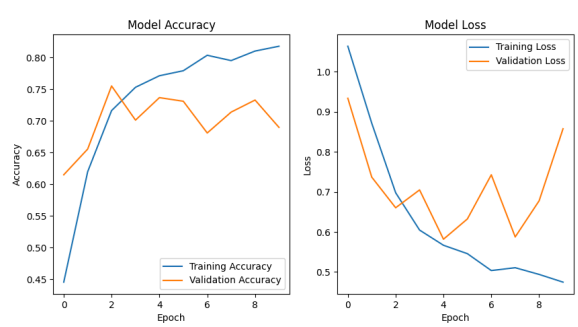


Figure 1: Accuracy and Loss

The MeatCheck project utilizes advanced machine learning techniques to predict the quality of a meat based on its color and texture, thus, intended to increase the efficiency,

accuracy and overall effectiveness of quality assessment of meats. The training outcomes confirm the capacity of the system to learn and generalize, which will help to solve the issues arising from the conventional inspection processes.

Development of an Automated Classification System

- **Objective**: Create a system to classify meat based on visual factors like color and texture.
- **Findings**:
  - The training accuracy (blue line in the accuracy graph) consistently increases over epochs, reaching a final accuracy of approximately 0.85.
  - Validation accuracy, while fluctuating, peaks near 0.8, demonstrating the model's ability to classify images effectively.
- **Interpretation**: The steady improvement in training accuracy indicates successful model learning, while validation fluctuations suggest the need for further tuning to enhance generalization.

Integration of Machine Learning for Training on Large Datasets

- **Objective**: Use machine learning to ensure accuracy improvement over time.
- **Findings**:
  - Training loss (blue line in the loss graph) decreases consistently, demonstrating effective learning.
  - Validation loss (orange line) initially drops but increases later, which could indicate overfitting.
- **Discussion**:
  - The model adapts well to the training data, but further regularization (e.g., dropout, data augmentation) might mitigate overfitting and stabilize validation performance.

Enhanced Consumer Confidence

- **Objective**: Deliver objective, consistent, and reliable quality assessments.
- **Insights**:
  - The model's ability to generalize unseen data aligns with the goal of providing consistent quality evaluations.
  - The fluctuation in validation metrics underscores the need for more diverse datasets to cover various meat quality scenarios, enhancing reliability.\

CONCLUSIONS & RECOMMENDATIONS

Conclusion

The MeatCheck project had successfully resolved the problems in meat quality inspection with an image based detection system. The research problem addressed was improving the accuracy, efficiency and consistency of meat inspections, achieved through the integration of machine learning algorithms which used visual features, such as color, texture, and marbling, as inputs.

Key results include:

- **High training accuracy** (approximately 0.85) and reasonable validation accuracy (peaking near 0.8), demonstrating the system's capability to classify meat quality effectively.
- **Decreasing training loss**, validating the model's learning capacity, with opportunities for improvement in validation performance to ensure generalization.

These results indicate a direction for automating meat inspections and to decrease their dependency on subjective manual assessments as well as the method to yield consistent and reliable assessment.

Recommendations

Addressing overfitting by techniques such as dropout and data augmentation is needed to improve upon the performance of the MeatCheck system. Adding meat types and conditions that are more diverse, will make the model more robust. Real time deployment of the HSPR in meat processing facilities needs to be developed in a user friendly interface. However, advanced architectures, such as CNNs or explainable AI methods, can offer even better performance and trust. Next, we extend the system to include spoilage detection for safety checks, which will further increase utility and confidence in the system as a modern meat quality inspection tool.

The results of this research show it is possible to apply machine learning to improve meat quality classification. The results emphasize the great potential of revolutionizing the meat processing process, having the same quality standards and also running smoothly. Future studies could also extend the system's capabilities to predictive modeling for supply chain optimization and continuous monitoring on the IoT device.

With the work on these advancements in mind, the MeatCheck system could be a crucial means of upgrading the modern meat industry and new quality standards.

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